Aakash Bartakke

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EDUCATION

Binghamton University, State University of New York — Binghamton, NY
Master of Science in Computer Science (AI Track) — *Aug 2024 – Dec 2025*
Dr. Babasaheb Ambedkar Technological University — Lonere, India
Bachelor of Technology in Computer Science — *Aug 2016 – May 2020*

EXPERIENCE

Senior Data Engineer — *Dec 2020 – Jun 2024*
LTIMindtree — *Pune, India*

- Architected cross-factory data warehousing platform for supply chain business, processing 10M+ daily records from 40+ sources, **replacing 8-10s QAD** reports with **sub-2 second refresh Power BI dashboards** for 20+ users, **unlocking $110K+ profit** in year one by enabling adequate product phase-out decisions and overstocking prevention.
- Engineered ESG reporting system processing 500K+ daily records through SQL, SSIS and BOM-to-emissions logic, **lowering tax liability by 14% ($2.2M+ annually)** by calculating per-product taxable waste to dispute government levy estimates.
- Optimized ETL pipelines across 10+ applications by implementing **delta load mechanisms and database tuning practices** (indexing, partitioning, query optimization), **reducing average runtime by 60-75%** maintaining 100% SLA adherence for 200+ business users.
- Eliminated **80% pipeline monitoring efforts** by building **automated dashboards** using Power BI displaying status, failures, and logs, replacing manual scripting across a 10-person support team.

Research Assistant — *Dec 2024 – Present*
Binghamton University — *Binghamton, NY*

- Developing **real-time data acquisition** system for thermal cooling research, integrating **200+ process sensors** across data center server racks using PyVISA instrument communication, replacing proprietary LabVIEW system with PySide6 GUI and PyQtGraph visualization, threshold-based alarms and structured logging to **eliminate high licensing costs**.

PROJECTS

NYC Taxi Analytics Platform | Databricks, Delta Lake, Snowflake, Spark, FastAPI, Spark MLlib, S3 — *Dec 2025 – Mar 2026*

- Engineered production-grade **Medallion Architecture** processing 200M+ historical plus 4M daily taxi trips using Databricks, **PySpark** and Delta Lake on S3, with idempotent ingestion and **sub-6-minute end-to-end load**.
- Designed analytics dashboards and **automated Snowflake warehouse ingestion** using Snowpipe and Streams, powering demand forecasting, surge pricing, and fleet optimization projected to deliver **8-15% revenue lift**.
- Built **ML-powered recommendation engine** for EV taxi drivers using H3 spatial indexing and XGBoost Trees, serving **682K+ precomputed recommendations** via FastAPI for data-driven repositioning to tackle deadheading.
- Built **spatial analysis tool** mapping EV charger coverage across high-demand zones using Streamlit dashboard with PyDeck demand heatmaps and charger overlay surfacing actionable infrastructure investment signals for fleet operators and city planners.

Real-Time Sales Ingestion Pipeline | Kafka, AWS EC2, S3, DynamoDB, Airflow — *Jun 2025 – Jul 2025*

- Engineered real-time data ingestion pipeline, streaming sales data across 5 countries through **Apache Kafka** on AWS EC2, with consumer batch processing and **Airflow**-orchestrated workflows, achieving **sub-$4/day cloud infrastructure cost,** targeted for small retail business entities.

TECHNICAL SKILLS

Data Engineering: Apache Spark (PySpark), Databricks, Apache Kafka, Apache Airflow, Delta Lake, SSIS
Cloud & Warehousing: AWS (S3, EC2, DynamoDB), Azure (Data Factory, Data Lake), Snowflake (Snowpipe, Streams, Tasks)
Languages & Tools: Python (Pandas, NumPy, Boto3), SQL (T-SQL, Stored Procedures), Java, Git, Docker, Linux, FastAPI
Analytics & Visualization: Power BI, Data Modeling (Star Schema, Snowflake Schema), ETL/ELT Patterns, Streamlit
Methodologies: Medallion Architecture, Incremental Processing, CDC, Data Quality, Agile/Scrum